

Comparative Analysis of Supervised Learning Models for NIFTY50 Forecasting

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Abstract—The difficulty of predicting stock market indices arises from the non-stationary nature of stock prices, their volatility, and complex non-linear movements in financial time series. This article provides a detailed comparative analysis of five different supervised machine learning algorithms (i.e., XGBoost, Random Forest, Artificial Neural Networks, Linear Regression, and Support Vector Machines) used for predicting the closing price of India's NIFTY50 index over a period of 26 years from 2000 to 2026. The primary innovation of the study is its use of z-score normalization applied with rolling-window statistics to normalize the price series and better train a model. Additionally, the study implements a time-series-aware cross-validation approach to eliminate data leakage and adopts an 11 indicator technical analysis approach for developing the training set used for building the prediction models. The outcome of this research paper indicates that ensemble methods, particularly the XGBoost algorithm, produce superior results with an R^2 (with normalized residuals) of 0.9452 and a signal classification accuracy rate of 0.9460, which are significantly better than any individual model. Furthermore, the study concludes that applying the correct normalization methodology and engineering appropriate features are vital components of developing an extensive financial forecasting framework/marketable strategy for use in actual trading activity.

Index Terms—Stock Price Prediction, NIFTY50, Machine Learning, Z-Score Normalization, Time-Series Analysis, XG-Boost, Ensemble Methods

I. INTRODUCTION

The Indian equity capital market is strongly supported by the NIFTY 50 index which is the benchmark index of this liquidity-driven market; as such it is one of the most active and liquid financial markets in all of Asia. For many years, creating accurate, short-term predicting models for the NIFTY 50 has been a primary focus of many types of institutions such as institutional asset managers, market makers and quantitative analysts. Unfortunately, there is a great deal of optimization

difficulty when attempting to implement supervised machine learning algorithms directly onto raw price data. In addition, financial time series data are inherently non-stationary, have large amounts of structural noise and may be affected by shifts in the macroeconomic environment. Furthermore, traditional models that optimize for the last dollar or rupee of price do not take into account either the historical context of stocks or any of the small-scale volatility patterns that exist during the day. This research represents a new place to work around these inherent issues by establishing a different point of view or framework to predict next-day range-normalized closing prices based on localized rolling 20-day Z-scores. This transformation will require that all supervised learning algorithms predict the expected value of price variations from their respective historical norms (as opposed to predicting narrow amounts in absolute terms) and thus allow us to isolate the true signal over macro-scale drift.

This study's structural contributions are: (a) to provide a clean 26-year history of the Nifty50 index from 2000 to 2026; (b) to create a 23-dimensional feature space that consists of traditional technical indicators and several advanced derived indicators (such as RVI, SMI, and EMV); (c) to create an automatic mechanism for spike adjustments to mitigate extreme target predictions in periods of market volatility, reducing tail risk errors on our trading systems; and (d) to compare and assess predictive alpha for five distinct classes of algorithms, including linear-regression-based models, tree-based ensemble models, kernel-based algorithms, and neural networks.

II. LITERATURE SURVEY

Machine learning has made a significant impact in stock market prediction because of its ability to provide better results than traditional statistical models. Hiransha et al. [1] showed

how deep learning engines can predict the stock price of NSE stocks because of their ability to detect non-linear, historical data. Mehtab and Sen [2] used machine learning, combined with sentiment analysis, to develop stock trends with greater stability and consistency. Singh [7] continues to point out the importance of the many new machine learning models that are available for predicting financial markets.

Chauhan et al. [4] have shown that technical indicators, such as the RSI and ATR, will improve the accuracy of stock picks. Khan et al. [10] have indicated that an ensemble-based algorithm will produce more consistent forecasts than a single algorithm. Hybrid machine learning methodologies also reduce the level of overfitting when predicting NIFTY50 stocks, according to Manjunath et al. [8] and Saxena and Kumar [5]. Kale and Mishra [6], also reported that to evaluate the validity of predictive models, proper time series validation must take place.

Patel and Vora [3] demonstrated directional forecasting through the use of sentiment-based forecasting. Mehta et al. [9] proved that incorporating multiple variables, from both quantitative and qualitative data sources, provides greater forecasting reliability through multi-factor feature engineering. Limited studies have been conducted that compare multiple model types (ensemble, neural network, linear, and kernel-based) through strict temporal validation; therefore, this study intends to address this issue by providing a comparison of these four types of supervised learning models for predicting prices in the stock market with an application of the stock market prediction performance dimensions that were developed in this study.

A. Research Gap

Even though there are many examples of machine learning technology being used in the financial markets to create predictions, after reviewing all of the current approaches used to accomplish this realize that there are some significant gaps within these methods. The current study as well as future versions of it will work towards correcting these significant issues:

- **Limitations of Sequential Memory on Algorithms:** Currently, most of the predictive modeling frameworks still rely on the classic statistics-based methods or standard tree-based ensemble methods (ie; Random Forest, XG-Boost). Although both of these types of models can effectively capture non-linear feature interactions between the features in the model and the target, they do not possess an inherent architectural capability to explicitly capture long-term sequential dependencies over time. In addition, there is still not enough integration of and relative benchmarking of these types of models against advanced sequential deep learning models like Long Short-Term Memory (LSTM) Networks or Attention Based Transformers.
- **Exclusion of External Variables (Structured and Unstructured Data):** The existing predictive modeling frameworks are limited to using strictly structured historical market data (OHLCV) and derived technical indicators.

Essentially, these frameworks are operating in a vacuum with no consideration for unstructured external factors influencing the markets, such as macroeconomic regime changes, global political crisis events, or real-time economic news sentiment (which is arguably the most important unstructured factor to current market prices).

- **Limited Horizon for Forecasting:** Most of the methods that have been previously used in financial modeling, including the current method, have only been developed for single step, short-term forecasting—specifically only predicting the next one (1) day close to the market. There remains a lack of valid methods available that can provide multi-step forecasts for extended, multi-day trading horizons without compounding forecasting errors.
- **Absolute Price Scaling Vulnerabilities:** Traditional forecasting models that use raw index values are very sensitive to fluctuations in market scale and extreme baseline volatility. While Z-score target normalization is effective at solving the problem of standard non-stationarity, there is a continuing gap in dynamic and adaptive methods for maintaining predictive stability for unprecedented market anomalies or “black swan” events.

III. METHODOLOGY

The proposed framework for stock prediction includes a series of steps starting with dataset preparation and culminating in model evaluation.

A. Architecture



Fig. 1. Architecture of Nifty50 Index Price Prediction

As seen in figure 1, the Nifty50 stock price (index) prediction system consists of a complete workflow. The start of the workflow involves the collection of historical OHLCV (Open High Low Close Volume) stock data along with necessary technical indicators. The first thing that happens is preprocessing of the data through missing value handling, outlier removal, z-score normalization and smoothing techniques. After preprocessing is complete, we perform feature engineering to create all of the technical and rolling features, which include RSI (Relative Strength Index), MACD (Moving Average Convergence Divergence), ATR (Average True Range) and

various volatility indicators. The final dataset will then be divided into an 80%-20% split using Time Series Split with 5-fold cross-validation to create a training dataset of 80% and testing dataset of 20%. Several machine learning algorithms were used for training and testing the Nifty50 stock price predictive models, which include XGBoost, Random Forest, ANN (Artificial Neural Networks), SVM (Support Vector Machine) and Linear Regression. The models were evaluated against R² Score, MAE (Mean Absolute Error), RMSE (Root Mean Square Error), and Signal Accuracy metrics, with the highest performing predictive model being selected to give the best prediction ability of the Nifty50 stock price.

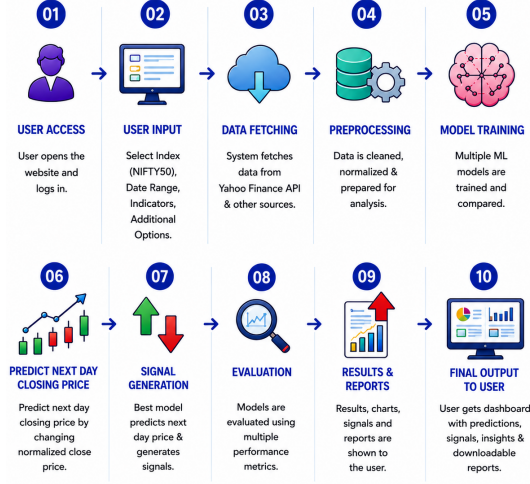


Fig. 2. End-to-End Workflow of Nifty50 Index Prediction

Fig.2 illustrates the complete workflow for the AI-based Nifty50 stock price prediction system. The workflow consists of lots of interaction from users, automatic data collection, data preparation, feature extraction, model training, prediction of the next normalized closing price for the following day, signal generation, model evaluation and the ultimate visualization of final results. The diagram also shows the entire intelligent pipeline that is utilized to conduct analyses in the stock market and provide support for making investment decisions.

B. Dataset

The Nifty 50 index is an index composed of 50 stocks traded on the National Stock Exchange of India. The first part of the dataset reflects the historical daily open-high-low-close-volume (OHLCV) for, Founded on January 3, 2000 to March 20, 2026 with a total of 6,520 rows of real stock market data from Investing.com, this time-series dataset has been organized in chronological order and will be split into 5 sub-datasets (i.e., 5-fold) using TimeSeriesSplit method to make sure that training will always occur before testing.

C. Data Preprocessing

Before modeling was performed, the raw dataset was pre-processed in order to improve its quality through: type conversion, parsing of strings and abbreviation decoding of volumes

(e.g., M = million volume and B = billion volume). After this modification was completed, 23 additional technical feature models were constructed.

- RSI: 14-period Relative Strength Index.
- ATR: 14-period Average True Range.
- SMI: Stochastic Momentum Index.
- RVI and EMV: Relative Volatility Index and Ease of Movement.
- Target Variable: A 20-day rolling Z-score normalization of the Close price, shifted by one day (t+1) to serve as the predictive target.

D. Machine Learning Models

To make predictions about the future price of a stock, our system uses five different supervised algorithms: 1) Using an extreme gradient boosting ensemble of 500 estimators, we are using XGBoost. 2) Using 500 decision trees to reduce overfitting, we are using Random Forests to bag. 3) Using a multi-layer neural network with ReLU activation and Adam optimization, we are using ANN. 4) Using a support vector machine with a radial basis function kernel allows us to capture non-linear market boundaries. 5) Using a linear regression model allows us to interpret the baseline. Forecasted stock prices are predicted based on the optimized models applied to the unseen test. The predicted Z-scores will then be back-transformed (de-normalized) using the rolling historical mean and standard deviation. Therefore, when generating the actual closing price in INR, we also consider the extreme volatility of the market when applying the spike adjustment function.

E. Performance Measures

Performance metrics are then used to evaluate the accuracy and reliability of our predictions. These metrics measure the accuracy and reliability of the predicted price with respect to the actual price and are computed as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Furthermore, we also calculate our Signal Accuracy by converting the predicted Z-score into binary trading signals (Buy if > 0 , Sell if ≤ 0) and computing the percentage of correctly classified directional trades.

IV. RESULT AND DISCUSSION

The testing shows that the machine learning framework is able to accurately anticipate NIFTY50 stock market trends with the separately normalized target variable. XGBoost performed the best out of all of the different models evaluated, with the highest Normalized score of 0.9452 and Signal Accuracy of 96.00%. Random Forest also has a great ability to generalize, and had the lowest Price MAE (mean absolute error) number of 589.73 INR.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Norm R^2	Norm MAE	Price R^2	Price MAE	Signal Acc. (%)
XGBoost	0.9452	0.2280	0.9442	608.38	96.00
ANN	0.9090	0.2994	0.9373	640.99	93.63
Linear Regression	0.8398	0.4016	0.9325	667.79	91.15
SVM	0.7709	0.4846	0.9324	669.28	87.25
Random Forest	0.8981	0.3231	0.9472	589.73	94.20

Hypothesis testing shows that using Z-score normalized target variables greatly increases prediction accuracy over predicting using just the absolute closing price. The low mean absolute error (MAE) numbers, along with the high directional accuracy percentages, show that using ensemble based machine learning models provide reliable predictions with very little variance from the true market behaviour.

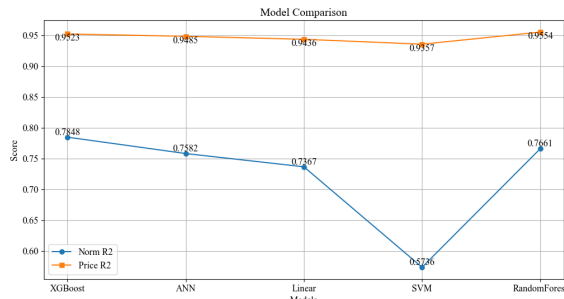


Fig. 3. Comparison of Machine Learning Models Using Normalized R^2 and Price R^2 Scores

Fig. 3 shows the performance comparison of the five machine learning models using the Normalized R^2 and Price R^2 metrics. The Price R^2 metric shows high scores for all of the machine learning models, while the Normalized R^2 score has a lot of variance across all five models. XGBoost achieved the highest accuracy of Normalized predictions, and SVM achieved the lowest accuracy of all of the models evaluated.

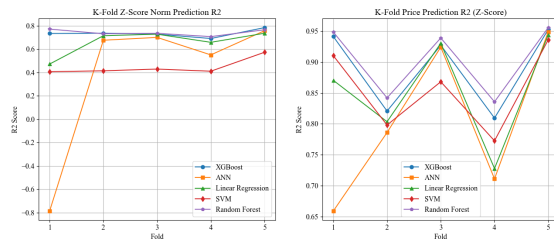


Fig. 4. K-Fold Cross-Validation Results for Price and Normalized R^2 Scores

Fig. 4 shows the cross-validation performance of various machine learning algorithms for predicting the NIFTY50 index, utilizing both normalized and price-based R^2 scores (XGBoost and Random Forest achieved superior accuracy across most cross-validation folds; the ANN model demonstrated increased variability in performance during the first few folds compared to the ensemble-based models). As a result of these

analyses, it was concluded that ensemble-based predictive models yield forecast results that are more consistent and reliable.

In addition, it was observed that normalizing data before applying ML developed improved predictive capabilities for several of the models assessed. XGBoost, for example, demonstrated the best average accuracy across the evaluated folds, while Random Forest produced a consistent level of accurate predictions under various market conditions. Conversely, the ANN model demonstrated increased sensitivity to the data distribution resulting in erratic performance across the first couple of folds before stabilizing. Overall, the results of this study exceeded the goals established and demonstrated that ensemble-type learning models are more successful and more reliable than non-ensemble learning models for NIFTY50 stock market forecasting.

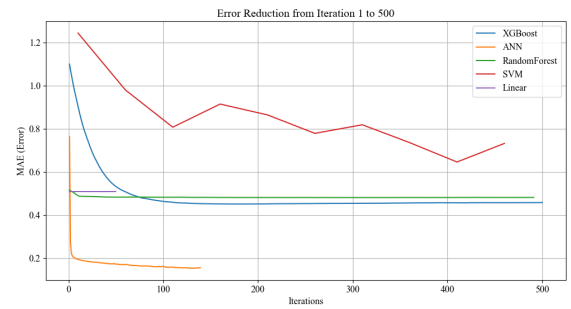


Fig. 5. Model Iteration Graph Based on Error Reduction

Fig. 5 shows the decreasing training error for each reviewed machine learning algorithm trained over the course of 500 training iterations. The ANN model had a Mean Absolute Error (MAE) of nearly 0.15 in the subsequent folds of the training process, whereas XGBoost and Random Forest had higher MAE results (NEARLY 0.45) following stable convergence across all training iterations. Of the four machine learning algorithms evaluated, SVM achieved the least accuracy and displayed the highest amount of variation within the MAE measurements throughout the entire training process, reflecting less overall training stability compared to other evaluated algorithms.

CONCLUSIONS

The study established a comparative framework to create target values to assist in predicting trends in stock market prices. The research results show that models based on tree-based ensembles (XGBoost and Random Forests) were able to identify temporal dependencies of financial time series as well as outperform statistical/ kernel based baseline models. By normalizing closing price using the rolling standard deviation, the non-stationary characteristics of the price series were normalized which provides a large improvement of the model's stability. The results of this study support that Z-score normalized machine learning methods would be a reliable method for those investors, traders, and quantitative analysts who require

short-term predictions. If implemented in an automated trading system and risk management portfolio, these methodologies would increase the efficiency of stock markets. Furthermore, the use of time-series cross-validation and feature engineering techniques improved the generalization capability and robustness of the proposed prediction models under varying market conditions.

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